Hard-sphere Colloidal Dispersions: The Scaling of Rheological Properties with Particle Size, Volume Fraction, and Shear Rate

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Synopsis

The steady-shear rheological properties of four submicron sterically stabilized silica dispersions differing in particle size were measured. The high- and low shear limiting viscosities were found to be a function of the volume fraction only and the volume fraction at which the viscosity diverges was found to be $\varphi_m = 0.63 \pm 0.02$ in the low shear limit and $\varphi_m = 0.71 \pm 0.02$ in the high shear limit, independent of particle size. The shear-thinning behavior scales in the Peclet number. At higher volume fractions, the shear-thinning transition shifts to a lower Peclet number. The time scale on which the shear-thinning transition takes place is comparable to the time scale of short-time self-diffusion, that is D_s^{short}/a^2 and relates to the structural changes in the dispersion which were observed in an earlier rheo-optical study.

I. INTRODUCTION

The rheological properties of colloidal dispersions are determined by an interplay of thermodynamical and fluid mechanical interactions. In a dispersion at rest, the particle distribution is determined by Brownian motion and direct potential interactions between the particles; the distribution functions are fully determined by thermodynamics. In a sheared suspension, the flow field disturbs the equilibrium particle distribution and then both thermodynamical and fluid mechanical variables determine the structure. This coupling of thermodynamical and fluid mechanical influences on the particle distribution prevents separate treatment of the thermodynamical and fluid mechanical sides of the problem. A satisfactory general microscopic description of the flow behavior of a dense suspension of interacting

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particles cannot be given yet. The problem can be simplified considerably by neglecting one side (or even parts) of the problem.

For instance, for hard spheres, namely spherical particles that do not interact directly except upon contact, several authors have described the viscosity in various limiting cases. Batchelor and Green[1–4] gave an exact description of the viscosity of a semidilute suspension of hard spheres in the limit of vanishing shear rate. This theory was extended to higher particle concentrations by Russel,[5] who took the mean thermodynamic force in a concentrated dispersion into account, but still treated the hydrodynamic interactions at the pair level.

Several other methods have been developed to calculate the viscosity of (hard-sphere) suspensions. Beenakker[6] calculates the effective viscosity using a formal solution for the hydrodynamic interaction between an arbitrary number of spheres, as developed by Van Saarloos and Mazur.[7] Beenakker gives an expansion for the viscosity in terms of fluctuations of the volume fraction around its average. Using the Percus-Yevick (equilibrium) correlation functions for hard spheres to describe the supramolecular fluid structure, he calculates the viscosity in second order of the expansion mentioned. This theory neglects Brownian motion. Note the Beenakker's calculations can and must be compared with viscosity measurements in which the particle distribution is only weakly disturbed and the Brownian motion effect is not present. These conditions are met in measurements of the high frequency limit of linear viscoelastic behavior.[8,9]

Bedeaux et al.[10] calculated the viscosity as a function of volume fraction taking the hydrodynamic interaction between an arbitrary number of spheres into account. They use a multipole expansion in the force density and also neglect Brownian motion.

Other authors, for example, Krieger and Dougherty[11], Saitô,[12] and Quemada[13] have given semiempirical or mean field descriptions for the viscosity as a function of shear rate and/or volume fraction; this has led to a variety of formulas.

Computers allow (limited) simulations of suspension dyamics to be made. In this respect the work of Bossis and Brady[14] on two-dimensional systems of hard disks is of interest. They include full fluid dynamical interactions in their simulations. It is interesting to compare molecular dynamics simulations on simple liquids by Woodcock,[15] Evans,[16] Hess,[17] and others to the phenomena ob-

served in colloidal suspensions. The main observation of all these authors is the structural ordering at higher shear rates.

On the experimental side there is a wealth of results. Burgers[18] made a compilation in 1962 and Jinescu[19] reviewed experimental data in 1974. Croucher and Milkie[20] selectively reviewed data concerning the rheological properties of sterically stabilized non-aqueous dispersions. A very broad review of the rheological properties of particulate suspensions was given by Kamal and Mutel.[21] Recently, Barnes et al.[22] discussed various experimental results together with results of computer simulations of dense suspensions. Although there are many data, it is still virtually impossible to sort out the effects of one variable on the observed rheological behavior. There are hardly any studies in which one parameter (e.g., volume fraction, particle size, particle interaction, particle shape, shear rate) is systematically varied while all others are kept contant, or even more important, are well defined. Since characterizing the interaction potential (independently, not as an adjustable parameter) is difficult, it is in this aspect that many studies are incomplete. An obvious choice for a model system with which to study suspension rheology is of course the hard-sphere colloidal dispersion. Again, it is difficult to establish the particle potential independently. However, this can be done by measuring the supramolecular fluid structure with light-scattering experiments, and by comparing it with the calculated structure factor S(K).[23] The osmotic compressibility over the full density range may also give a clue to the nature of the interactions.[24]

The purpose of this article is to present rheological data concerning a well-defined colloidal suspension. In Section II we shall discuss the ideas of Batchelor and Russel and some of the molecular dynamics results for dense suspensions. In Section III viscosity measurements will be dealt with. In the last section we shall discuss the obtained results.

For our study we used a hard-sphere colloidal dispersion. Although hard-sphere colloidal dispersions are not frequently encountered in practice, we think that the hard-sphere model can serve as a basis for obtaining an understanding of more complicated systems.

In a previous report,[25] one of our group described the volume fraction dependence of the low and high shear viscosity of a ster-

ically stabilized silica dispersion as a function of the volume fraction. It was found that the low shear viscosity diverges at $\varphi = 0.63$ and the high shear viscosity at $\varphi = 0.71$. Frequency dependent (low shear) measurements on these silica systems show a viscoelastic behavior in the frequency range of 80 Hz to 190 kHz.[8,9] The relaxation mechanism can be explained in terms of the interplay of shear and Brownian motion.

Our model system consists of undeformable submicron silica particles dispersed in cyclohexane, sterically stabilized by octadecyl chains grafted on to the particle surface. Light-scattering studies show that the thermodynamic behavior of these dispersions follows the hard sphere theory up to high volume fractions.[24] Recent measurements of the static structure factor S(K) by neutron scattering show that the structure factor corresponds to the structure factor predicted for hard spheres up to a volume fraction of $\varphi = 0.50$.[23] The hydrodynamic interactions between the particles in the limit of low volume fraction also conform to theories developed by Einstein,[26] and Stokes and Batchelor[27] for the viscosity and sedimentation velocity.

We present here measurements on the viscosity of four hard-sphere suspensions differing in particle size as a function of shear rate and volume fraction. It will be shown that scaling behavior is present as predicted by theory. In addition we shall discuss the agreement between various theoretical predictions and the measurements.

II. THEORETICAL BACKGROUND

A. Constitutive Equation

The viscosity η of a fluid relates the rate of strain, which is characterized by the rate of strain tensor $\dot{\mathbf{E}}$, to the deviatoric part of the bulk stress needed to maintain this rate of strain. The constitutive equation reads:

$$\sum = -P\mathbf{I} + 2\eta(\dot{E})\dot{\mathbf{E}} \tag{1}$$

where P is the hydrostatic pressure, Σ the stress tensor, and $\mathbf{I}$ is the unit tensor. The rate of strain dependence of η indicates the

general non-Newtonian behavior of the viscosity of suspensions. For simple shear flow Eq. (1) reduces to

$$\tau = \eta(\dot{\gamma})\dot{\gamma} \tag{2}$$

where the shear stress is $\tau = \Sigma_{21}$ and $\dot{\gamma} = 2\dot{E}_{21}$ is the shear rate.

B. Dimensional Analysis

Krieger[28] showed that the relative viscosity of a suspension of hard spheres, that is, the viscosity of the dispersion relative to the viscosity of the solvent, depends on five dimensionless groups in general. From this analysis follows that the steady-state shear viscosity depends on only two of these groups. The relative viscosity is determined by the volume fraction of the particles, which is defined as $\varphi = 4/3\ \pi n a^3$, where n is the number density of the particles and a is the particle radius. The second determining group is the Peclet number: $\mathrm{Pe} = a^2\dot{\gamma}/D_0 = 6\pi\eta_s a^3\dot{\gamma}/k_B T$ where D_0 is the Stokes-Einstein diffusion coefficient, η_s is the solvent viscosity, and $k_B T$ is the thermal energy. This number expresses the ratio between the time scale on which the structure of the dispersion is deformed by the shear and the time scale of Brownian motion that restores the structure.

C. Dilute and Semidilute Suspensions

The viscosity of a dispersion of hard, undeformable spheres was given by Einstein up to the first order in volume fraction in 1906/1911. He calculated the extra energy dissipation in the fluid due to the presence of the noninteracting particles.[26] This classical result is given by

$$\eta/\eta_s = 1 + (5/2)\varphi \tag{3}$$

where

η = viscosity of the dispersion
η_s = viscosity of the suspending fluid
φ = volume fraction of hard spheres

Up to order φ, the viscosity of a dispersion is a single-particle problem and the viscosity is Newtonian. Non-Newtonian effects

arise when particle interactions are taken into account and when the shear affects these interactions.

In slightly more dense suspensions, the particles can no longer be regarded as independent. The disturbance of the flow field near a particle due to the presence of a second particle leads to a φ^2 term in the expansion of the viscosity in the volume fraction.

Batchelor calculated the bulk stress due to the hydrodynamic interactions between pairs of spheres in a linear flow field. His expression for the bulk stress is given by[3,5,8,25,29]

$$\sum^H = 2\eta_s\dot{\mathbf{E}}\left(1 + \frac{5}{2}\varphi + \frac{5}{2}\varphi^2 + \frac{15}{2}\varphi^2 \int_2^\infty J(\rho)g(\rho)\rho^2\,d\rho\right) + I.T. \tag{4}$$

where $I.T.$ denotes an isotropic term. ρ is a dimensionless distance, $\rho = (r/a)$, where a is the particle radius. $J(\rho)$ represents the near-field hydrodynamic interactions. The expression for Σ^H involves the radial distribution function $g(\rho)$. This implies that the hydrodynamic part of the bulk stress is influenced by Brownian motion and direct interactions, since both affect $g(\rho)$. This is the so-called indirect contribution by Brownian motion and direct pair interactions to the bulk stress. However, for a dilute suspension in a weak flow the distribution function is written as

$$g = 1 + \mathrm{O(Pe)} \tag{5}$$

where we assumed that $\mathrm{Pe} \ll 1$. Therefore, this indirect contribution can be neglected. Batchelor and Green[3] calculated (using $g = 1$)

$$\sum^H = 2\eta_s\dot{\mathbf{E}}\left(1 + \frac{5}{2}\varphi + 5.2\varphi^2\right) + I.T. \tag{6}$$

This result can be regarded as the bulk stress in a high frequency low shear oscillatory flow because it applies to small deformations and does not allow for Brownian motion.

Apart from its indirect influence, the Brownian motion contributes directly to the bulk stress. For hard spheres, the direct contribution of Brownian motion is given by[4,5,8,25,29]

$$\sum^B = \frac{9}{20}\eta_s\varphi^2\dot{\mathbf{E}} \int_2^\infty \rho^2 W(\rho)g(\rho)f\,d\rho + I.T. = 2\eta_s\dot{\mathbf{E}}(0.97)\varphi^2 + I.T. \tag{7}$$

and the total bulk stress in a semidilute suspension of hard spheres is given by

$$\sum = \sum^{H} + \sum^{B} = 2\dot{\mathbf{E}}\eta_s(1 + 2.5\varphi + 6.2\varphi^2) + I.T. \tag{8}$$

and thus

$$\eta_r(\mathrm{Pe} \rightarrow 0) = 1 + 2.5\varphi + 6.2\varphi^2 \tag{9}$$

For a complete review of the influence of the colloidal forces on the rheology of suspensions we refer to an excellent review by Russel.[30]

D. Dense Suspensions

The theory outlined above is valid for dilute systems, where pair interactions are of interest only. Russel and Gast[5] extended this theory to higher volume fractions (but still at low shear rates) by taking the full thermodynamics into account. At higher volume fractions, the radial distribution function can no longer be set equal to one and the hard spheres generate a potential of mean force which is related to $g(r)$ as

$$V_{mf} = -k_B T \ln g(r) \tag{10}$$

This potential of mean force is thus taken as acting as an interaction potential between pairs of spheres. The hydrodynamic interactions are still treated at the pair-interaction level. The theory describes the divergence of the viscosity at high volume fractions qualitatively, but there is a discrepancy between the calculated results and the experimental data because the hydrodynamics are treated on the pair-interaction level.

Although Beenakker's[6] result cannot be compared with the results of steady shear measurements, we note that his calculations of the high frequency limit of the real part of the complex viscosity agree with our measurements of the complex viscosity of the hard sphere silica systems SP23, SSF1, and SJ18 up to a volume fraction $\varphi = 0.45$.[9]

Knowledge about the microstructure in dense suspensions can be obtained from computer simulations. Bossis and Brady[14] describe suspension dynamics experiments on a two-dimensional system of hard disks; they neglect Brownian motion, but take hydrodynamic interactions into account. They find a distortion

of the local particle order, particle clustering, and particle ordering. Woodcock[15] applies damped Newtonian molecular dynamics to dense systems. His calculations, which do neglect hydrodynamic particle interactions, show amorphous order at low shear rates and layering phenomena at high shear rates. At moderate shear rates (the shear-thinning region) both phases coexist.

Recently, we performed a rheo-optical study on system SJ18 up to volume fractions of 0.55.[31] It was found that at low shear rates the structure factor S(K) was deformed due to the shear. The characteristic time scale on which the ordering occurs coincides with the characteristic times observed for the shear thinning behavior (Section IV.E), illustrating the important effect of the distorted microstructure on the rheological properties of a dense suspension. At very high shear rates, we found no evidence that a new random structure appears, causing shear thickening as was suggested by Barnes et al.[22] Simulations by Heyes on a Lennard-Jones liquid[32] indicate that the shear rate at which shear thickening is first observed depends on the rate of change of the shear rate up to that point. The longer the time the system has to accommodate the shear strains by structural reorganization, the higher is the shear rate at which the shear thickening occurs.

III. EXPERIMENTAL

A. Particle Characterization

The colloidal model system consists of sterically stabilized monodisperse silica particles suspended in cyclohexane. The particles are synthesized as described by Stöber,[33] whereupon a stabilizing layer is grafted on to the particle surface. In the first step, tetraethoxysilane and an aqueous solution of ammonia are added to ethanol. Hydrolysis of the tetraethoxysilane catalyzed by the ammonia produces tetrahydroxysilane monomers, which polymerize to spherical particles. In this synthesis, the radius of the synthesized particles depends on the relative concentrations of the reactants. In the second step, octadecyl-alcohol ($C_{18}H_{37}OH$) is added to the dispersion of charge-stabilized silica particles. Then ethanol, water, and ammonia are distilled off from the re-

action mixture which yields a dispersion of silica particles in fluid octadecyl-alcohol. During this process, the environment of the particles changes gradually from polar to nonpolar and a monolayer of octadecyl chains is grafted chemically on to the surface. Finally, most of the excess octadecyl-alcohol is removed by vacuum distillation. The residue is dissolved in cyclohexane and further purified by repeated centrifugation. A detailed description of the preparation method is given by Van Helden et al.[34]

We prepared four dispersions containing particles of different sizes. The hydrodynamic radius (from dynamic light scattering, DLS) of the particles is shown in Table I, together with the radii found from a Guinier plot (static light scattering, SLS). The degree of polydispersity is indicated by the parameter σ, which is the (relative) standard deviation in the mean particle radius as was determined with electron microscopy.

The dispersion coded SP23 was also used for small angle neutron-scattering measurements,[23] for DLS measurements in benzene as a function of temperature[35] and for linear viscoelastic measurements.[8,9] The system coded SJ18 was already used in steady shear viscosity measurements,[25] linear viscoelastic measurements,[8,9] and rheo-optical measurements.[31] SSF1 was also used for linear viscoelastic measurements.[8,9] We mention this so that results of different studies can be correlated. More important however, is the fact that the "regeneration" for different experiments shows the intrinsic stability of the dispersions. Now and then we have to synthesize a new system. This involves considerable work in synthesis and characterization. Results of experiments do not depend on batch number.

B. Sample Preparation

Samples for the capillary viscometer (Ubbelohde type) measurements were prepared as follows. The concentration of the

TABLE I
Radii of the Silica Particles

System	Radius (nm) (DLS)	$(a/a_{SP23})^3$	Radius (nm) (SLS)	σ
SP23	28	1	28	0.12
SSF1	46	4.4	48	0.13
SJ18	76	20.0	77	0.09
SJ14	110	60.6	115–130	0.12

stock dispersion was determined by drying a known volume of the dispersion in a nitrogen stream at 60°C. The sample was dried for several hours until all the solvent was evaporated (no change of the dry weight with time). The weight concentration was always determined in duplicate and always agreed within 2%. Determinations by different persons led to the same result. Given the mass concentration and their specific volume (see Section III.E, one can calculate the volume fraction of the silica particles in the dispersion. Then 10 mL. of the stock dispersion was pipetted into the Ubbelohde viscometer. The samples can be diluted in the Ubbelohde viscometer reservoir. The dimensions of the Ubbelohde viscometer allow for a tenfold dilution of the stock sample without it having to be removed from the reservoir.

Low concentration samples for couette and parallel-plate measurements were prepared by diluting a fairly concentrated dispersion in a measuring flask. The concentration was determined by drying a known volume of the stock dispersion. Higher concentrations were prepared by evaporating cyclohexane from an amount of sample of known concentration, volume, and density. A flask (of known weight) containing the sample was placed in a water bath in 60°C, and dust-free nitrogen was gently blown over the mouth of the flask until the required weight was reached. The concentration was checked once more by drying a small quantity of concentrated sample. The highest concentrations were prepared by weighing an amount of dried silica in a measuring flask. Cyclohexane was added and the flask was reweighed. It is not possible to pipette these concentrated samples because a slight evaporation of cyclohexane from the tip of the pipette causes blockage.

C. Rheometers

We used two types of rheometers, an Ubbelohde capillary viscometer (Schott-Gerate GmbH, Germany) and a Deer rheometer (Deer Ltd., England). Both viscometers are microcomputer controlled. The Deer rheometer can be equipped with a couette as well as a parallel-plate measuring cell designed and built in our laboratory. Both geometries contain a vapor-lock system which prevents evaporation of the solvent from the sample. Figure 1 shows the couette geometry with vapor-lock system. Details of both geometries are described elsewhere.[25]

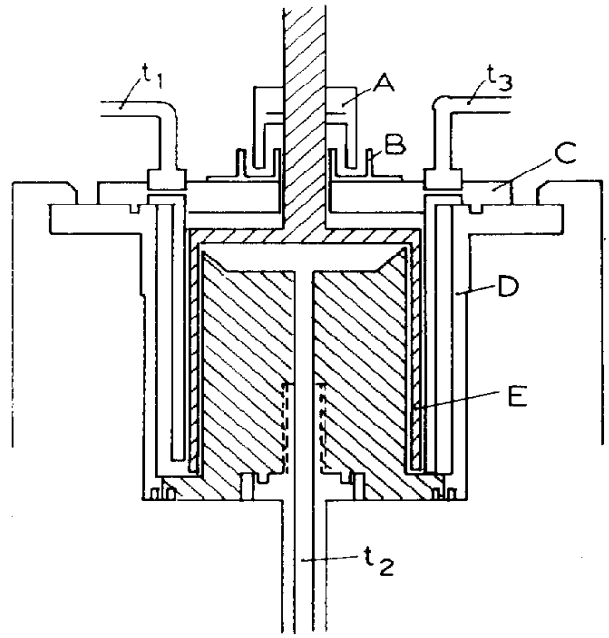

Fig. 1. Couette geometry as used for the viscosity measurements. A-Teflon manchet; B-water trough; C-glass coverplate: D-stainless steel cup; E-stainless steel bob; t_1 and t_2-filling and cleaning lines; t_3-thermometer insert. The (inner) diameter of the outer cup is 4.43 cm.

The Deer rheometer is a constant stress apparatus: a constant torque, which is a measure of the shear stress depending on the measuring geometry is applied to the measuring cell and the constant speed, which is proportional to the shear rate, is measured. The rheometer can operate in the torque range $0 < \tau < 10^{-2}$ Nm. or in the speed range 0–40 rad s^{-1}. Here, we report measurements in which we used only the couette cell. All measurements we performed with the same cell. The gap width is 0.50 mm. We previously reported[25] measurements on system SJ18 using the parallel-plate cell as well as the couette cell. In the measurements with the parallel-plate cell the gap was varied between 50 μm and 150 μm. The data obtained with both cells are consistent and we therefore are confident that no wall effects are present. The rheometer was calibrated using calibration oils obtained from TNO-Zeist.

D. Measuring Procedure

Ubbelohde

A thoroughly cleaned Ubbelohde viscometer is filled with cyclohexane and placed in a water bath maintained at a temperature of 20.00 ± 0.01°C. For the measurements of the specific volume (Sect. III.E) we used a capillary with a radius R = 0.26 mm. The time t required to empty a volume V of 5 mL is typically 250 ± 0.03 s. The representative shear rate $\dot{\gamma}_{rep} = V/(tR^3) = 1150\ s^{-1}$. For measurements on more concentrated samples of SP23 we used a capillary with a radius R = 0.39 mm. (see Sect. IV). Evaporation from the sample is suppressed by connecting the viscometer to a vessel containing cyclohexane.

First, cyclohexane is measured at least 10 times. The cyclohexane is removed from the reservoir and the viscometer is dried in air stream. A small amount of sample (10 mL typically) is pipetted into the Ubbelohde viscometer and measured 10 times. Then a certain volume of cyclohexane is added to the dispersion and the contents of the Ubbelohde reservoir are homogenized. The diluted dispersion is measured. More dilutions are made in the same manner until the reservoir is filled completely (in our case the maximum volume is 100 mL). After the dispersion has been removed from the reservoir, the Ubbelohde is rinsed with cyclohexane. Finally the cyclohexane is remeasured.

Deer Rheometer

The measuring chamber is rinsed thoroughly with cyclohexane and then dried in an airstream. Then the vapor-lock system is filled with water and a blank run measures the residual friction of the air-bearing and the vapor-lock system. Before each measurement on a silica sample, typically ten runs on cyclohexane are performed as a reference. In every run, 40 regularly spaced values are taken in the interval $\omega = 0 - \omega = 40$ rad s^{-1}. After the cyclohexane has been removed and the cell has been dried with an airstream, the measuring chamber is filled with a silica dispersion. Again, at least 10 runs are made. Typically, 40 points are taken between 0 and 40 rad s^{-1}, or in the interval $\tau = 0 - 10^{-2}$ Nm, depending on which limit is reached first. If necessary, the data can be concentrated in or spaced out over a region of interest.

The viscosities determined from the torque-speed relation contain calibration factors which can be eliminated by scaling the viscosity to the apparent viscosity of cyclohexane. Furthermore, one has to correct for a contribution to the apparent viscosity from the friction in the air bearing of the rheometer. The latter can be found from the τ versus ω relation in air. However small, this correction is important, especially at low particle concentrations. The relative viscosity at a certain $\dot{\gamma}$ is thus calculated as

$$\eta_r(\dot{\gamma}) = \frac{\eta'_{sa}(\dot{\gamma}) - \eta'_a}{\eta_c - \eta_a}$$

where the prime denotes the appearent values found from the τ-ω dependence and the subscripts sa, c, a refer to sample, cyclohexane and air, respectively. Typical values for the calibration factors are $\eta'_a = 10$, $\eta'_c = 186$, independent of shear rate.

E. Specific Volumes

The specific volumes were determined with the Ubbelohde capillary viscometer. The specific volume q can be defined as $q = \varphi/c$, where φ is the volume fraction and c is the (weight) particle concentration. These silica particles behave as hard spheres, therefore, they follow Einstein's correction on the viscosity in the low concentration limit [Eq. (3)]. In this limit, the specific volume can be determined by applying Eq. (3) to measurements of η_r versus c. The viscosity of dilute dispersions ($c <$ 0.025 g cm^{-3}) was measured and the specific volume was obtained by fitting $\frac{2}{5}(\eta_r - 1)$ versus c. The slope of the plot gives the specific volume q as is shown in Figure 2 for the systems SP23 and SSF1. The results are summarized in Table II.

In view of the uncertainty in the weight concentration of the samples (see Sect. III.B) and the uncertainty in the specific volume, the uncertainty in the volume fraction is less than 4%.

It could be argued that the specific volume should be determined from an independent experiment such as dynamic light scattering. However, in such an experiment another moment of the size distribution is measured. If this factor is taken into account, then results of different experiments will yield the same specific volume. For small particles there is a slight tendency toward an interaction specific volume (e.g., from osmotic compressibility measurements) that is smaller than the hydrodynamic specific

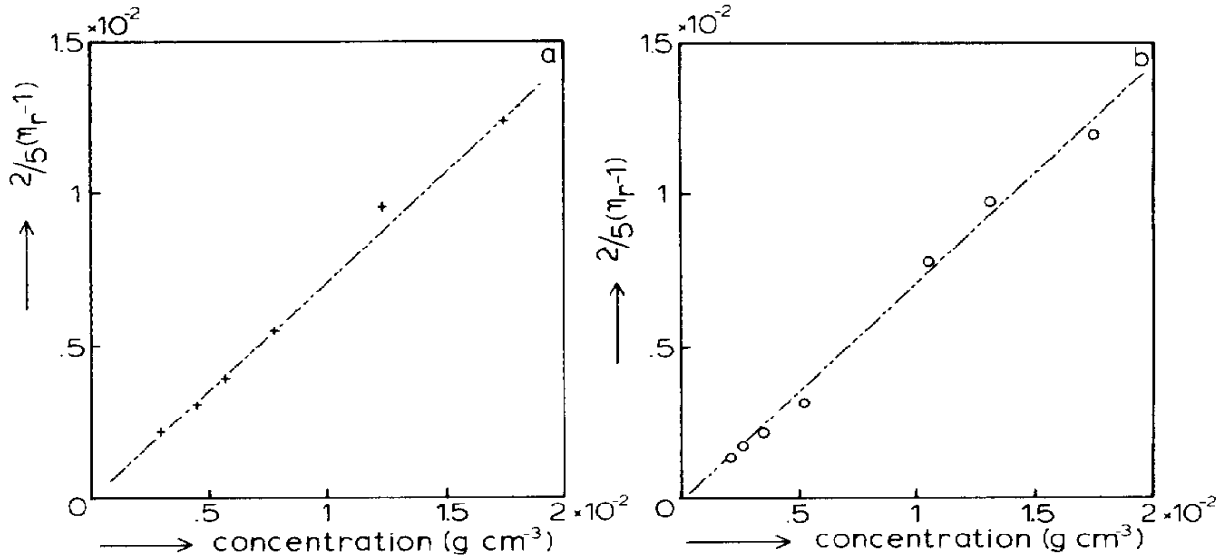

Fig. 2. Specific volume of systems SP23 (a) and SSF1 (b) from Ubbelohde measurements at low particle concentration.

volume. We think that the use of Einstein's equation gives a hydrodynamic volume of a "single" particle. The use of this value should give consistent results for all volume fractions up to close packing.

F. Data Reduction

The Deer rheometer essentially provides arrays of the nominal applied torque τ versus the speed of the rotating cylinder ω, which is a measure of the shear rate $\dot{\gamma}$. Using the appropriate calibration constants, the viscosity η can in principle be calculated from every data pair. To find the limiting viscosities how-

TABLE II
Specific Hydrodynamic Volume of Colloidal Silica Particles Dispersed in Cyclohexane

System	$q/(\mathrm{cm^3 g^{-1}})$
SP23	0.71 ± 0.02
SSF1	0.69 ± 0.02
SJ18	0.63 ± 0.02
SJ14	0.65 ± 0.02

ever, it often is necessary to fit the data and extrapolate to $\dot{\gamma} = 0$ or $\dot{\gamma} = \infty$, depending on which limiting viscosity one wants to determine. Of course, such an extrapolation will yield the correct value for the viscosity only if the measurements are done near the limits. If an apparent Newtonian behavior was shown the data were fitted linearly. The slope is then a measure of the viscosity.

System SP23 was measured in the couette cell in a reduced shear rate range $1 \times 10^{-4} < \mathrm{Pe} < 0.2$ at the lower concentrations. Shear-thinning effects could be seen only at the highest shear rates, but the high shear limit was not reached. From these measurements the low shear viscosity was calculated by fitting linearly in the Newtonian region. The results shown in Figure 3 are typical.

The system SSF1 containing the particles with a 46 nm radius was measured with the couette cell. The data were reduced as described for system SP23.

The low and high shear data from system SJ18 have already been presented in great detail.[25] Measurements were done with both couette and parallel-plate geometries. At low and moderate concentrations ($\varphi < 0.45$) couette measurements could be done in the Peclet range $10^{-2} < \mathrm{Pe} < 5$. From these data the low shear

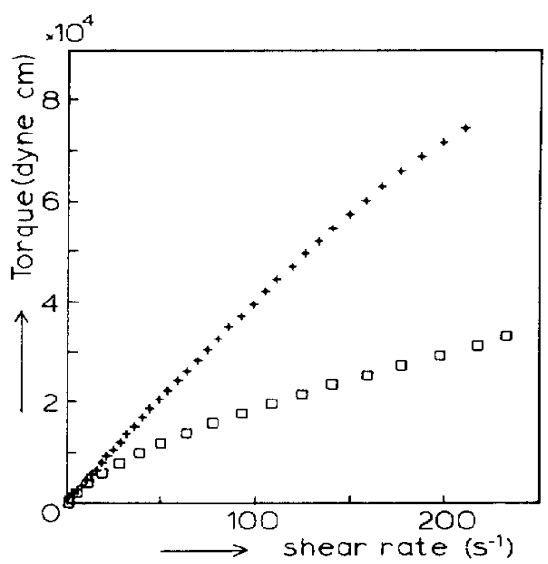

Fig. 3. The shear stress vs. shear rate for system SP23 at $\varphi = 0.558$ (+) and for SJ14 at $\varphi = 0.556$ (□).

limiting viscosity could be determined by fitting the data to the equation

$$\tau = a_1\dot{\gamma} + a_3(1 - e^{-(a_2/a_3)\dot{\gamma}}),$$

this equation is purely empirical but was found to describe the data adequately over a large shear rate range. The low shear viscosity is then determined by $\eta(\dot{\gamma} = 0) = a_1 + a_2$, and the high shear limiting value could be estimated using $\eta(\dot{\gamma} = \infty) = a_1$.

In view of the range of shear rates covered, one may well doubt whether the high shear region was reached. The obtained values however were found to be in accordance with data obtained with the parallel-plate geometry, which allowed much higher shear rates ($\mathrm{Pe}_{\max} = 100$).

Couette measurements on system SJ14 ($a = 110$ nm), in the range $1 \times 10^{-2} < \mathrm{Pe} < 10$ provided values for $\eta(\dot{\gamma} \to \infty)$ upon fitting linearly in the high shear region. Low shear values could be extrapolated with reasonable accuracy. At the highest volume fractions ($\varphi > 0.4$), the uncertainty is fairly large. Figure 3 is a typical example of τ versus ω for system SJ14.

IV. RESULTS AND DISCUSSION

A. Comparison of Viscosities Measured with Different Types of Rheometers

Before comparing the results obtained with different types of rheometers, one must be aware of the following points. First, the nature of the flow must be the same in both rheometers so that the perturbation of the pair distribution function (the nonequilibrium microstructure) is the same in both experiments. Second, the results can only be compared at the same shear rate: the rate at which the particle distribution is perturbed also determines the viscosity.

The set of rheometers we used does not satisfy the first condition. The couette geometry provides a simple shear flow, whereas in a capillary viscometer a Poiseuille flow is established. Figure 4 compares data for the system of the smallest particles used, SP23, obtained with both the Deer rheometer with couette geometry and the Ubbelohde capillary viscosimeter. The comparison must be made at the representative shear rate

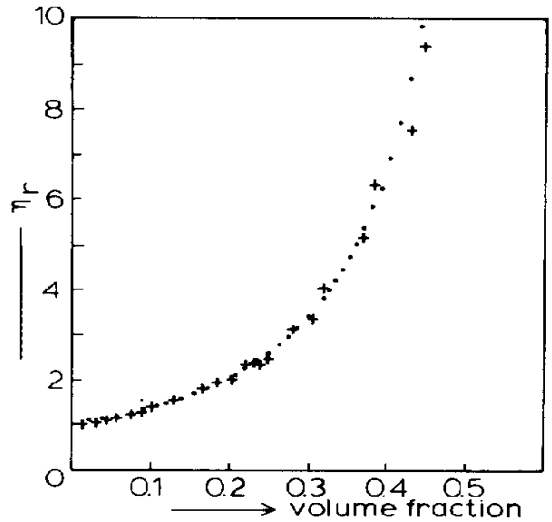

Fig. 4. Comparison of the values of the low Peclet viscosity of SP23 measured with two types of rheometers. (.) Ubbelohde viscosimeter. (+) Deer rheometer with couette geometry.

of the Ubbelohde viscometer, which depends on the viscosity of the sample. For these measurements we used a capillary of radius R = 0.39 mm. For a fairly concentrated sample, (φ = 0.35, η = 4 mPas) the representative shear rate $\dot{\gamma}_{rep}$ = 400 s^{-1}. Although this shear rate seems rather high, the Peclet number is still low, Pe = 0.04. At this Peclet number, this sample shows no shear thinning. Therefore, we used the $\eta(\dot{\gamma} = 0)$ values. Up to φ = 0.35, both sets of data coincide completely. At higher volume fractions, the viscosities measured with the Ubbelohde tend to exceed the values determined with the Deer rheometer. This can be due to the difference in shear flow in the rheometers.

The discrepancy, if any, cannot be due to the fact that we compare with $\eta(\dot{\gamma} = 0)$ values and not with $\eta(\dot{\gamma}_{rep})$ values.

B. Relative Viscosity as a Function of Volume Fraction and Particle Size

Figure 5 shows the (low shear) viscosity for systems of different particle size as a function of volume fraction up to φ = 0.35. The data are found to scale within the experimental error in the volume fraction and no particle size effect is seen. Batchelor's

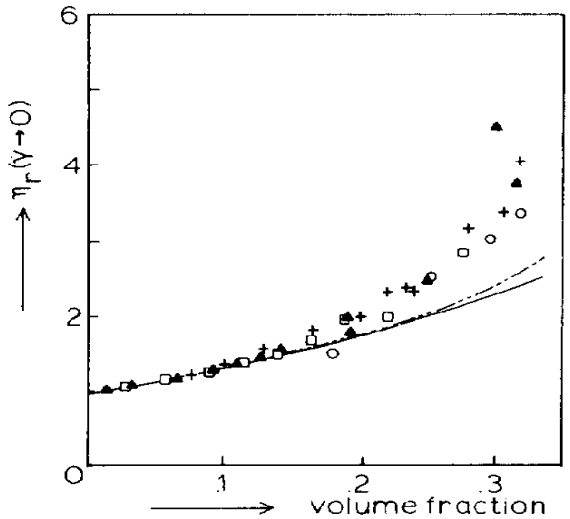

Fig. 5. Low shear viscosities of SP23 (+), SJ18 (▲), SSF1 (○), and SJ14 (□) as a function of volume fraction. Solid line: Batchelor theory. Dashed line: Russel and Gast theory. Although it is not shown in this figure, note that the Russel and Gast theory describes the strong increase of the (low shear) viscosity with volume fraction near the maximum packing fraction qualitatively.

results describe the volume fraction dependence adequately up to $\varphi = 0.1$. Above this volume fraction, the equilibrium radial distribution function can no longer be set to unity, which leads to a potential of mean force which acts as an effective pair potential that enhances the stress. Russel and Gast[5] performed calculations on the nonequilibrium pair-distribution function (in the low shear case) thereby emphasizing the thermodynamic side of the problem, but still assuming pairwise additivity of all interactions. The dashed line in Figure 5 represents the result they found to describe the strong increase of the viscosity qualitatively at higher volume fractions. There is still a discrepancy between the data and the prediction, obviously due to the incomplete treatment of the hydrodynamics.

The relative viscosity as a function of volume fraction for the four silica systems is given in Figure 6 in the range $0 < \varphi < 0.60$. For the two smallest particle systems, only the low shear viscosity is measured reliably. For the two larger particle dispersions, both low and high shear limiting viscosities are plotted. The solid line is according to Quemada's[13] semiempirical expression for the volume fraction dependence of the relative viscosity, viz.

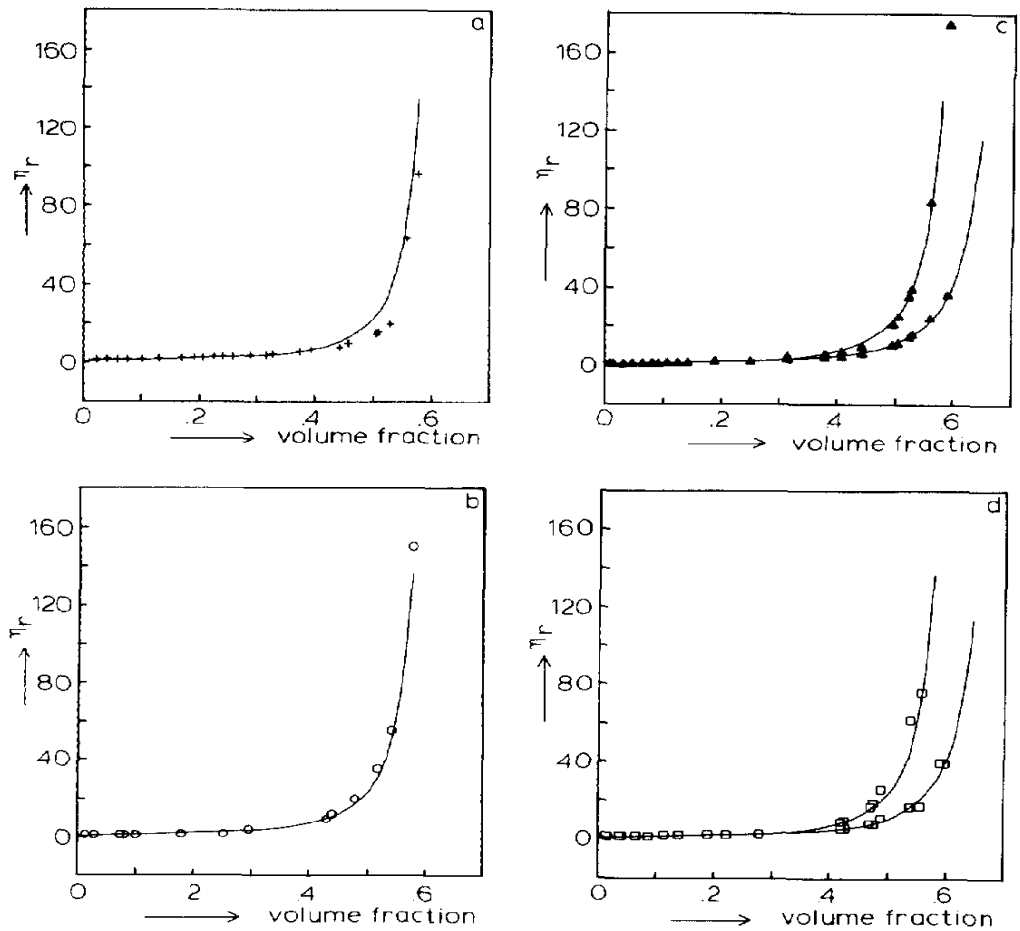

Fig. 6. Relative viscosity as a function of volume fraction of four dispersions containing different sized spheres. Dashed lines according to Eq. (11). In all figures the same values for φ_m were used. In the limits $\dot{\gamma} \to 0$, $\varphi_m = 0.63$, $\dot{\gamma} \to \infty$, $\varphi_m = 0.71$. (a) SP23 a = 28 nm; (b) SSF1 a = 46 nm; (c) SJ18 a = 76 nm; (d) SJ14 a = 110 nm. The uncertainty in the volume fraction is less than 4%.

$$\eta_r = (1 - \varphi/\varphi_m)^{-2} \tag{11}$$

where φ_m is the maximum packing fraction. For all systems we used $\varphi_m = 0.63$ in the low shear limit, and $\varphi_m = 0.71$ in the high shear limit. Over the whole range of volume fractions, there is no particle size effect present and a perfect scaling in the volume fraction is observed. Within the experimental error, the viscosities seem to diverge at the same maximum packing fraction. In Figure 7 we have plotted $1/\eta_0$ and $1/\eta_\infty$ as a function of volume fraction. The data are given numerically in Tables III–VI.

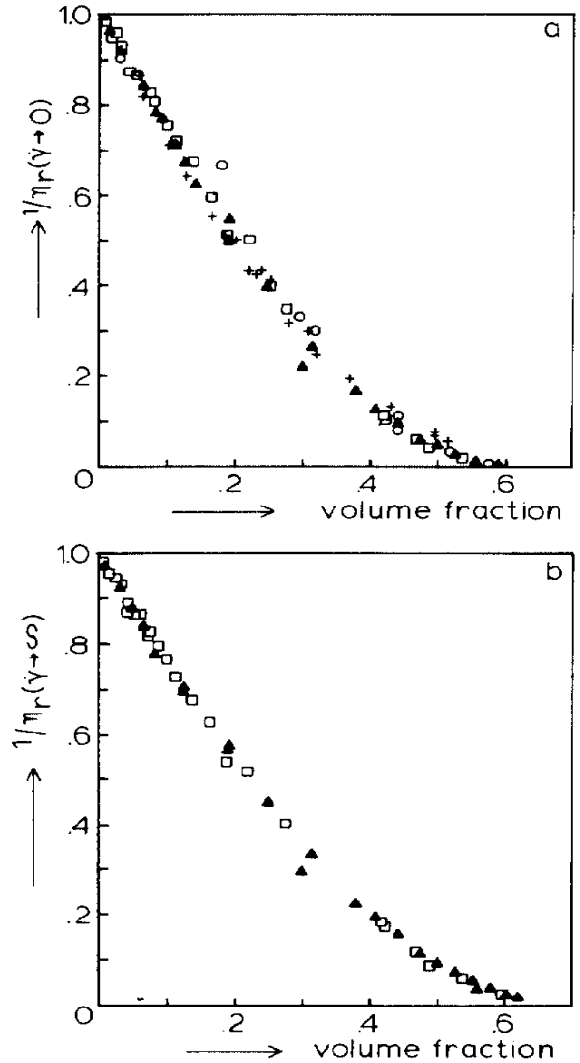

Fig. 7. The reciprocal shear viscosity as a function of volume fraction. SP23 (+), SSF1 (○), SJ18 (▲), and SJ14 (□). (a) low shear viscosities, (b) high shear viscosities.

C. Maximum Packing Fraction

We used Eq. (11) to find a value for the maximum packing fraction. It should be noted that if the Krieger-Dougherty equation[11] is fitted to the data of SJ18,[25] we get the same exponent and values for the maximum packing fractions as in Eq. (11). From Figure 6 it seems that Eq. (11) is a good description of the volume dependence of η_r at all volume fractions.

Bedeaux[36] suggested another representation of the viscosity data, namely an extension of a formula given by Saitô.[12] In Saitô's calculation, interactions and correlations between two or more spheres are neglected. To the Saitô formula Bedeaux adds a term $S(\varphi)$ which accounts for these contributions. This leads to the expression:

$$\frac{\eta_r - 1}{\eta_r + \frac{3}{2}} = \varphi[1 + S(\varphi)] \tag{12}$$

In Figure 8 the term $S(\varphi)$ is plotted for the dispersions of SJ18 and SJ14 together with the prediction made by Eq. (11) using the

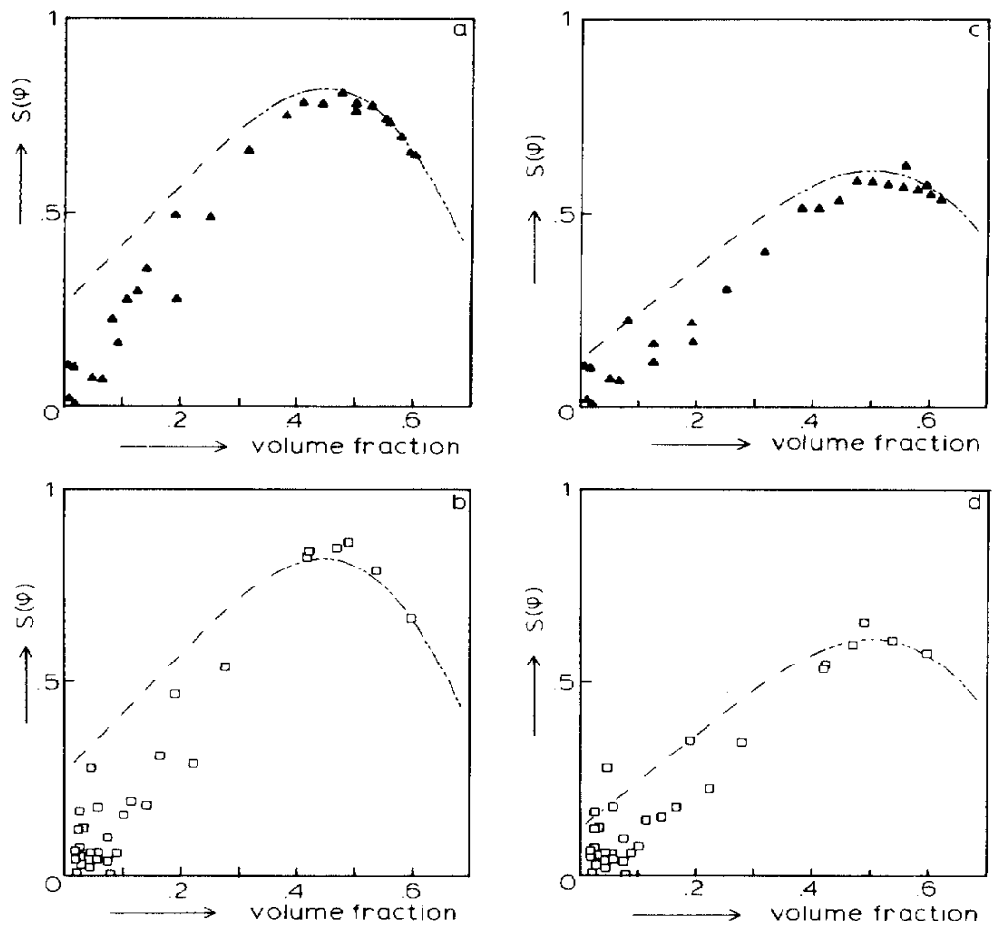

Fig. 8. Representation of the relative viscosity as suggested by Bedeaux.[36] The drawn line is obtained from Eq. (11) with $\varphi_m = 0.63$ in the low shear limit and in the high shear limit $\varphi_m = 0.71$. (a) SJ18 (low shear); (b) SJ14 (low shear); (c) SJ18 (high shear); (d) SJ14 (high shear).

values for φ_m mentioned above. The discrepancy between the data points and the modeled $S(\varphi)$ behavior at low φ clearly shows the incorrect value for the intrinsic viscosity found from Eq. (11) in which we used $\varphi_m = 0.63$ or 0.71. But Figure 8 clearly illustrates that the behavior of $S(\varphi)$, and therefore the behavior of η_r, is very well described at higher values of φ by the equation used. It is also for this reason that we think our extrapolated values for φ_m are adequate.

The value of φ_m is a function of shear rate. This phenomenon is thought to originate in the ordering effect of the shear flow. Since ordered suspensions can be filled more dense than disordered ones, it is not surprising that φ_m increases with increasing shear rate.

It seems logical to correlate the zero shear rate value $\varphi_m = 0.63$ to the random dense packing (rdp). From several experiments with millimeter-sized spheres as reviewed by Do Ik Lee,[37] an average value for φ_m(rdp) = 0.639 was found. It is interesting to note that this value was found after "tapping" the container. Without shaking or vibrating the container, the (millimeter) spheres pack loosely and the resulting value for φ_m in this random loose packing is φ_m(rlp) = 0.589.

We assume that the mechanical vibrations yield a higher φ_m as a result of the surface roughness of the macroscopic spheres. When two spheres touch, they must move laterally when they move radially. Applying mechanical vibrations to such a system clearly enables small lateral movements and leads to a denser packing. In our opinion, this view explains why dispersions of macroscopic spheres show a divergence of the viscosity at volume fractions φ_m (rlp) = 0.58 or lower; the shear pushes the particles together and immobilizes the system.

For colloidal systems, with particles smaller than 1 μm, the situation is different. Here the Brownian motion effectively "lubricates" the lateral movements and the viscosity seems not to diverge before $\varphi_m = 0.63$.

We observed that at higher shear rates the system seems capable to flow at volume fractions $\varphi > 0.63$. Shear-induced ordering can explain this observation. Suppose that the system crystallizes (locally) to a body-centered cubic (bcc) structure with the diagonal plane parallel to the flow direction. This situation is mechanically unstable, in other words, the system flows. An even more

convincing argument is the following observation, which to our knowledge was not made before. Suppose one has a close-packed crystal. The crystal contains hexagonal layers of spheres stacked as abab... or abcabc..., respectively, known as hexagonal close packed (hcp) or face-centered cubic (fcc) crystals. It can easily be calculated that the packing fraction of such a system equals $\pi/(3\sqrt{2}) = 0.7405$ for both hcp and fcc. The lattice spacing of such a crystal is $d\sqrt{(2/3)}$ where d is the sphere diameter. Suppose that the shear gradient is perpendicular to these layers and that we dilute the system so that the layer spacing is d. Now the layers can slide over each other. In this case the volume fraction $\varphi_m = (d\sqrt{(2/3)}/d) * 0.7405 = 0.605$. The layers can be packed even more densely when we allow them to slide in a zig-zag motion. Then the layer spacing can be small as $\frac{1}{2}d\sqrt{3}$. The corresponding packing fraction becomes $\varphi_m = (2/9)\pi = 0.70$, remarkably close to our high shear value $\varphi_m = 0.71 \pm 0.02$.

The above discussion implies that the shear field introduces order. Light-[31] and neutron-[38] scattering experiments on dense colloidal dispersions indeed show ordering with increasing shear rate (at Pe > 1). We expect that shearing dense dispersions at higher and higher Peclet numbers introduces more and more order leading to a limiting behavior in which hexagonal layers slide over each other in a zig-zag fashion up to $\varphi_m = 0.70$.

D. Particle Correlations at Higher Volume Fractions

The representation of the values of the high shear viscosity in Figure 8 shows another interesting feature, as was recognized by Bedeaux[36]. The slope of the plot of $S(\varphi)$ versus φ is virtually independent of φ up to volume fractions of $\varphi = 0.40$. The value of the slope is 1.2, which is close to the theoretical value of 1.14 that was calculated by Schmitz[39] taking only two particle correlations into account. Since the slope is independent of the volume fraction it might be possible, if the shear-distorted distribution functions in a dilute suspension are known, to calculate the high shear viscosity up to $\varphi = 0.40$.

E. Scaling with Peclet Number and Characteristic Shear Stress

In Figure 9 we plotted the characteristic Peclet number, as a function of volume fraction. This characteristic Peclet number

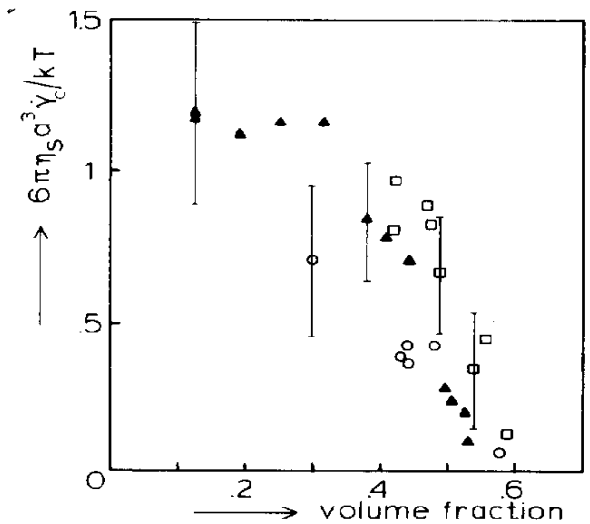

Fig. 9. The characteristic Peclet number $6\pi\eta_s a^3\dot{\gamma}_c/kT$ as a function of volume fraction. SSF1 (○), SJ18 (▲), SJ14 (□).

is defined as the Peclet number at which $\eta_r(\dot{\gamma}_c) = \eta_\infty + 1/2(\eta_0 - \eta_\infty)$; this Peclet number is a characteristic point on the shear-thinning curve η versus $\dot{\gamma}$. The characteristic Peclet number is independent of particle size. The characteristic shear rate scales with a^3/kT within experimental error.

To evaluate the characteristic shear rate for system SJ14, one has to make an assumption about the value of η_0. In those cases that we could not extrapolate a value of η_0 from the measurements, we estimated this value from Eq. (11), using $\varphi_m = 0.63$. The uncertainty in the volume fraction is 2–4% and above $\varphi = 0.40$ this leads to large uncertainties in the estimated values of the zero shear limiting viscosity.

The uncertainties in the values of $\dot{\gamma}_c$ of system SSF1 are introduced mainly by the estimate of η_∞ from Eq. (11). This uncertainty is much smaller than the one made in estimating η_0, since η_∞ varies less with volume fraction below $\varphi < 0.60$. Note that we made assumptions only about the values of the limiting viscosities. We did not make any assumption about the form of the shear thinning curve. All the characteristic shear rates in Figure 9 were measured experimentally. Despite the uncertainties involved, one can conclude that the characteristic Peclet number does depend on volume fraction but is independent of particle size within experimental error.

It is interesting to note that the time scale on which the shear thinning occurs parallels the time scale on which structural changes in system SJ18 were observed when light scattering techniques were used in a shear flow.[31] We found that a distorted fluid phase was transformed into a more ordered phase when a shear rate corresponding to the shear thinning region was applied. The exact nature of the ordering could not yet be identified. The shear thinning behavior is thus connected with the deformation of the microstructure occurring under the applied shear.

In Figure 10 it is shown that the shear-thinning behavior scales with the reduced characteristic shear stress $a^3\tau_c/kT$, where $\tau_c = \eta(\dot{\gamma}_c) * \dot{\gamma}_c$. This representation does not contain more information than the characteristic Peclet number, but we show this reduced characteristic shear stress in order to compare with experiments by Choi and Krieger.[42]

Choi and Krieger[42] show a scaling of the viscosity in the reduced shear stress, $\tau_r = a^3\eta(\dot{\gamma})\dot{\gamma}/kT$, for volume fractions $0.30 < \varphi < 0.45$. This reduced shear stress is effectively a Peclet number in which the diffusion constant D_0 is replaced by an effective diffusion constant D_{eff}. By replacing the solvent viscosity by the effective viscosity, one can hope to correct for the slow-down of the diffusion in concentrated dispersions. At low Peclet numbers

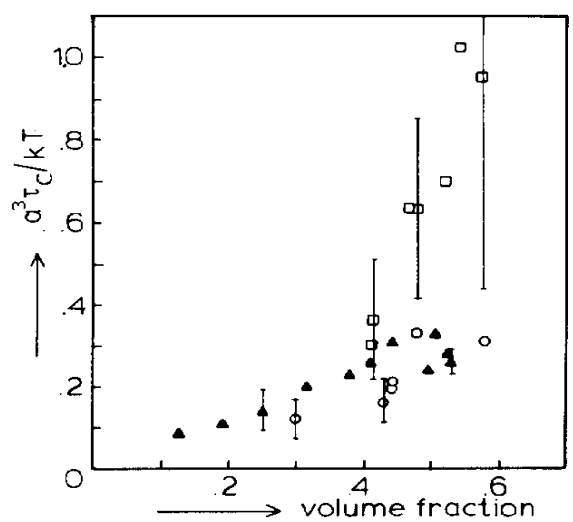

Fig. 10. The reduced characteristic shear stress, $a^3\tau_c/kT$ as a function of the volume fraction. SSF1 (○), SJ18 (▲), SJ14 (□).

and for volume fractions $\varphi < 0.25$ we argue that this correction is appropriate. We expect that the diffusion constant of interest closely connected to the short-time self-diffusion coefficient D_s^{short} (see also Ref. [9]). Every microscopic distortion of the particle distribution is counteracted by individual particle motions and the structure of the dispersion does not change. There can be no macroscopic gradient in the system to drive a collective diffusion process. It is known from both theoretical[40] and experimental[41] studies that for $\varphi < 0.25$, D_s^{short} does increase proportional to the inverse low shear (relative) viscosity. In Figure 11 we plot the relative viscosity calculated from experimental data for the short-time self-diffusion coefficient. For this we use

$$\eta_r = D_0/D_s^{\text{short}} \tag{13}$$

where D_0 is the diffusion coefficient at infinite dilution. We compare this result with the low shear viscosity of SJ18. The self-diffusion was measured in a disperson of sterically stabilized silica particles (a = 440 nm.) in tetrahydrofurfuryl alcohol.[41] At volume fractions $\varphi > 0.25$, the short-time self-diffusion coefficient does not decrease proportional to the decrease of the inverse low shear viscosity.

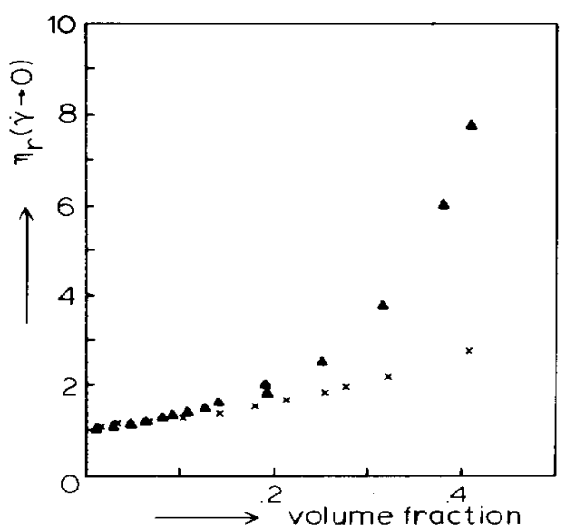

Fig. 11. Comparison of the low shear viscosity of system SJ18 (▲) and the relative viscosity calculated from the short-time self diffusion coefficient D_s^{short} (X).

TABLE III
Experimental Low Shear Viscosities of System SP23

c (g cm^{-3})	φ vol. frac	η_0
0.0203	0.014	1.038 ±.002
0.0253	0.018	1.054
0.0334	0.024	1.063
0.0410	0.029	1.084
0.0478	0.034	1.096
0.0583	0.041	1.098
0.0632	0.045	1.137
0.0699	0.050	1.147
0.0813	0.058	1.168 ±.002
0.0818	0.058	1.169 ±.002
0.0905	0.064	1.223
0.109	0.077	1.250
0.144	0.102	1.31 ±.01
0.185	0.131	1.41
0.239	0.170	1.81
0.269	0.191	1.96
0.290	0.206	2.01
0.319	0.226	2.33
0.338	0.240	2.37 ±.02
0.346	0.246	2.34
0.363	0.258	2.44
0.406	0.289	3.16
0.442	0.314	3.39
0.461	0.327	4.05 ±.05
0.534	0.374	5.17
0.554	0.393	6.33
0.622	0.442	7.5 ±.1
0.645	0.458	9.4
0.717	0.509	14.4
0.720	0.511	15.5
0.745	0.529	19.5
0.786	0.558	63 ±1
0.812	0.577	96
0.837	0.594	197 ±3

Using the reduced shear stress as a variable, Choi and Krieger[42] found a reduced characteristic shear stress $a^3\tau_c/kT$ independent of volume fraction in the volume fraction range $0.30 < \varphi < 0.45$. Indeed in this volume fraction range our data do not show a strong dependence of the reduced characteristic shear stress on the volume fraction (see Fig. 10). Over the whole

TABLE IV
Experimental Viscosities and Characteristic Values for System SSF1

c (g cm^{-3})	φ vol. frac	η_0	η_∞	η_c	Pe_c	$a^3\tau_c/kT$
0.0217	0.015	1.055 ±.002				
0.0437	0.030	1.107				
0.109	0.075	1.219				
0.117	0.080	1.240				
0.147	0.101	1.335				
0.262	0.181	1.509				
0.367	0.253	2.530				
0.430	0.297	3.77 ±.02	{3.0 ±.3}	3.4 ±.2	0.70 ±.24	0.12 ±.05
0.62	0.43	9.67	{6.4}	8.1	0.38	0.16
0.64	0.44	11.6	{6.9}	9.3	0.42	0.21
0.641	0.442	12.4 ±0.1	{7.0 ±1}	10 ±.5	0.36	0.19 ±.05
0.695	0.480	19.5	{9.5}	14.5	0.42	0.33
0.752	0.519	35.1	—	—	—	—
0.784	0.541	55.2	—	—	—	—
0.836	0.577	150 ±5	{29}	90	0.06	0.30

The bracketed values are calculated from Eq. (11), using φ_m=0.71.

TABLE V
Experimental Viscosities and Characteristic Values for System SJ18

c (g cm^{-3})	φ vol. frac	η_0	η_∞	η_c	Pe_c	$a^3\tau_c/kT$
0.0010	6.3×10^{-4}	1.004 ±.002	1.004			
0.0049	3.1×10^{-3}	1.006	1.006			
0.0102	6.5×10^{-3}	1.018	1.018			
0.0147	9.3×10^{-3}	1.024	1.024			
0.0199	1.3×10^{-2}	1.032	1.032			
0.0254	0.0160	1.045	1.045			
0.0299	0.0190	1.049 ±.005	1.049			
0.0507	0.0320	1.082	1.082			
0.0765	0.0483	1.137	1.137			
0.104	0.0655	1.188	1.188			
0.130	0.082	1.280	1.280			
0.146	0.092	1.300	—			
0.171	0.108	1.401	—			
0.199	0.126	1.49 ±.02	1.41 ±.02	1.45 ±.02	1.2 ±.4	0.09 ±.03
0.200	0.126	1.48	1.43	1.45	1.18	0.09
0.225	0.142	1.60	—	—	—	—
0.303	0.191	2.00	1.76	1.88	1.12	0.11
0.398	0.251	2.50	2.22	2.36	1.16	0.14
0.500	0.316	3.77	2.99	3.37	1.16	0.20
0.601	0.380	5.98 ±.1	4.39 ±.1	5.2 ±.1	0.84 ±.2	0.23 ±.04
0.648	0.409	7.76	5.09	6.4	0.78	0.26
0.701	0.443	10.35	6.32	8.3	0.70	0.31
0.787	0.496	22 ±2	11 ±1	17 ±2	0.28	0.24 ±.02
0.800	0.506	26	12	19	0.24	0.33
0.83	0.525	36	15	26	0.20	0.28
0.84	0.53	40	16	28	0.10	0.26
0.89	0.56	84 ±6	25 ±2	—	—	—
0.94	0.59	175 ±5	37	—	—	—

Most of these data were already reported in Ref. 25. For reasons of clarity we did not include parallel-plate measurements.

TABLE IV
Experimental Viscosities and Characteristic Values for System SJ14

c (g cm^{-3})	φ vol. frac	η_0	η_∞	η_c	Pe_c	$a^3\tau_c/k_BT$
0.0134	0.0085	1.020	1.020			
0.0235	0.015	1.036	1.036			
0.0273	0.018	1.048	1.048			
0.0307	0.020	1.047	1.047			
0.0335	0.022	1.043	1.043			
0.0380	0.025	1.074	1.074			
0.0380	0.025	1.071	1.071			
0.0427	0.028	1.073	1.073			
0.0480	0.031	1.091	1.091			
0.0488	0.032	1.077	1.077			
0.0570	0.037	1.082	1.082			
0.0683	0.044	1.118	1.118			
0.0854	0.056	1.156	1.156			
0.0961	0.062	1.157	1.157			
0.114	0.074	1.22	1.22			
0.119	0.077	1.21	1.21			
0.136	0.088	1.26	1.26			
0.175	0.114	1.39	1.37			
0.215	0.140	1.50	1.48			
0.291	0.189	1.96	1.86			
0.340	0.221	1.99	1.93			
0.426	0.277	2.85	2.49			
0.644	0.419	9.0 ±1	5.5 ±.2	7.2 ±.6	0.8 ±.4	0.3 ±.2
0.650	0.423	9.7	5.7	7.70	0.96	0.36
0.722	0.470	17.1	8.5	12.8	0.88	0.56
0.730	0.475	19.2	8.0	13.6	0.82	0.56
0.752	0.488	26 ±2	11.5 ±1	18.8 ±1.5	0.66 ±2	0.63 ±0.3
0.827	0.538	62	16.9	39.5	0.34	0.69
0.855	0.556	{73 ±5}	17.5 ±2	46	0.44 ±.2	1.0 ±.5
0.904	0.588	300[a]	40	170	0.12	0.95

The bracketed values are calculated from Eq. (11) using φ_m=0.63.
[a]This low shear value was extrapolated from the η vs. $\dot{\gamma}$ curve.

volume fraction range however, we see a dependence of the reduced characteristic shear stress on the volume fraction.

F. Possible Evidence for the Hard-Sphere Transition

In a previous paper[25] we showed that the characteristic shear stress, defined as the shear stress at which $\eta_r(\dot{\gamma}) = \eta_\infty + 1/2 \cdot (\eta_0 - \eta_\infty)$, is dependent on the volume fraction. It was shown that this quantity increases monotonically up to $\varphi = 0.5$. For $\varphi > 0.5$, the characteristic shear stress diminishes again. We noted that it is tempting to correlate this abrupt change in τ_c at $\varphi = 0.50$ to the hard-sphere liquid-solid transition, because the shear thinning behavior, in particular the viscosity at low shear rates, is very sensitive to changes in the equilibrium microstructure. The results of systems SSF1 and SJ14 as shown in Figure 10 neither contradict nor support this conclusion. However, both systems are more polydisperse than SJ18 (see Table I). Barrat and Hansen[43] and McRae and Haymet[44] showed that the hard-sphere liquid-solid transition "disappears" at too high polydispersities.

G. Absence of Shear Thickening Behavior

It should be noted that in our study we did not see any effect of shear thickening, even at the highest concentrations. It is not clear whether this absence is a fundamental property of our model system, or whether this phenomen will be present at higher Peclet number. Hoffman[45] suggested that at high Peclet numbers the colloids would order into hexagonal layers. At very high Peclet numbers the break-up of these layers would cause shear thickening. Although we have evidence for the formation of these hexagonal layers, we did not see any effect of this proposed break-up. In measurements on a silica suspension of 170 nm. radius particles (thus at $Pe_{max} = 40$) we found no evidence of shear thickening. Parallel-plate measurements on SJ18 allowed shear rates up to Pe = 100. No shear thickening was observed.

V. SUMMARY

We measured the shear viscosity of sterically stabilized colloidal dispersions of different sized silica particles (radius a = 28–110 nm) as a function of particle size, shear rate, and volume

fraction. The volume fraction of the samples was varied from 10^{-4} to 0.60. Using these different sized particles we were able to study the viscosity at Peclet numbers ranging from Pe = 10^{-4} to Pe = 10. In previous light- and neutron-scattering studies it was found that up to a volume fraction $\varphi = 0.50$ the interactions between the particles can be described very well by the hard-sphere potential.

The low and high shear limiting viscosities were found to be independent of particle size. The low shear limiting viscosity diverges at $\varphi_m = 0.63 \pm 0.02$ and the high shear limiting viscosity was found to diverge at $\varphi_m = 0.71 \pm 0.02$. The latter value for φ_m is well above the value for random dense packing. We argued however that dense-ordered packings can exist that permit flow at a packing fraction $\varphi = 0.70$. At a given volume fraction, the viscosity scales in the Peclet number.

We followed Krieger in characterizing the shear thinning curve by a characteristic Peclet number, namely the Peclet number at which $\eta_r(\mathrm{Pe}_c) = \eta_\infty + {}^1/_2(\eta_0 - \eta_\infty)$. This characteristic Peclet number is a function of the volume fraction only. The results are in qualitative agreement with data obtained by Choi and Krieger,[42] but are extended over a much larger volume fraction range.

In a previous report,[25] we reported that the characteristic reduced shear stress (which is another representation of the characteristic Peclet number) increases monotonically up to $\varphi = 0.50$. For $\varphi > 0.50$, the characteristic reduced shear stress diminishes again. We noted that it is tempting to correlate this abrupt change in τ_c to the hard-sphere liquid-solid transition. The new and additional data reported here neither contradict nor support this conclusion.

We finally note that we did not observe shear thickening. In a recent experiment no shear thickening was observed at Peclet numbers up to 1200.

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